Lab-7 Ex7 - Happy Numbers

Have you heard of the Happy Numbers? The origin of Happy Numbers is generally unknown but some say it came from Russia. Anyway, let's find out how happy Happy Numbers are?

- https://en.wikipedia.org/wiki/Happy_number#10-happy_numbers

To find out whether a positive number `x` is happy or not, we will do the following:

```
Step 1. Check if the number x is either 1 or 4. If it is 1, it is Happy.

If it is 4, it is Unhappy. Otherwise, continue to step 2.

Step 2. Replace the number x with the sum of the squares of its digits.

Go back to step 1.
```

For instance, for `x = 19`, we will follow the above method step by step (^2 denotes the square):

```
19
1^2 + 9^2 = 82
8^2 + 2^2 = 68
6^2 + 8^2 = 100
1^2 + 0^2 + 0^2 = 1
```

So we see that 19 is a Happy Number! Pretty obvious 100 is a Happy Number too.

[image] *"The Happy Children" on Wikipedia. Sorry, I couldn't find a photo for "Happy Numbers"*

Please write a program to get a **positive integer** `x` from user and check for happiness of the number step by step and **print out intermediate results**, as shown in the samples below. Please be reminded that we have previously learned to find out sum of digits using looping and the modulus operator (Lecture notes 05b).

Assume user input is always valid. User input is ***bold and italic***.

Sample run #1:

`x?` ***100***

`=> 1`

`Happy Number! :)`

Sample run #2:

`x?` ***4***

`Unhappy Number! :(`

Sample run #3:

`x?` ***103***

```
=> 10 => 1

Happy Number! :)
```

Sample run #4:

`x?` ***102***

```
=> 5 => 25 => 29 => 85 => 89 => 145 => 42 => 20 => 4

Unhappy Number! :(
```